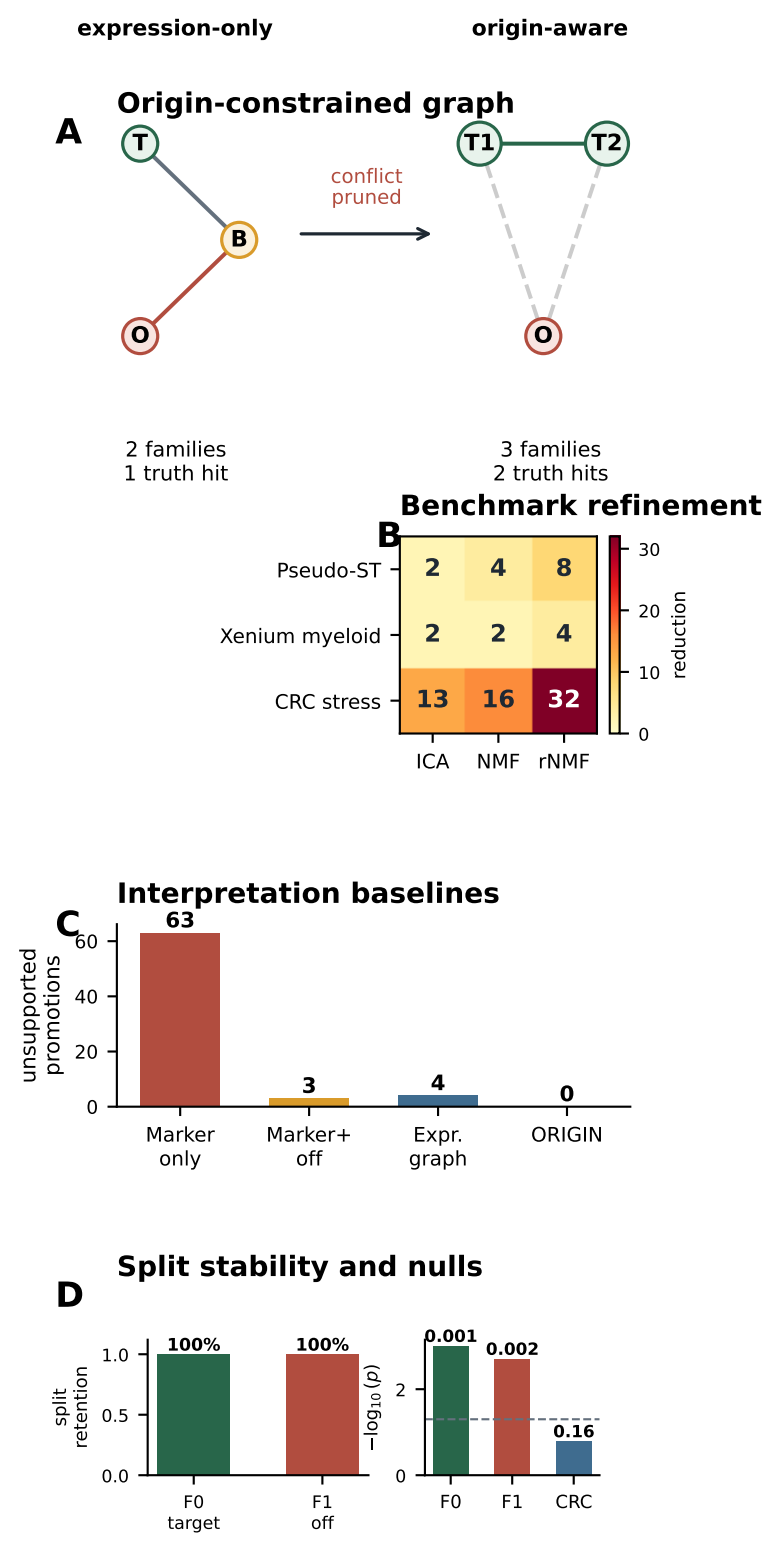
expression-only
origin-aware
A
Origin-constrained graph
T
B
O
conflict
pruned
T1
T2
O
2 families
1 truth hit
3 families
2 truth hits
B
Benchmark refinement
Pseudo-ST
Xenium myeloid
CRC stress
2
4
8
2
2
4
13
16
32
ICA
NMF
rNMF
30
20
10
0
reduction
C
Interpretation baselines
unsupported
promotions
63
3
4
0
Marker
only
Marker+
off
Expr.
graph
ORIGIN
D
Split stability and nulls
split
retention
100%
100%
F0
target
F1
off
$-\log_{10}(p)$
0.001
0.002
0.16
F0
F1
CRC